This document specifies the complete keyword library, decision logic, scoring rules, and batching protocol for the LungFirst AI flagging plugin. The tool reads structured and unstructured fields within Kanta patient records, applies a weighted scoring model, and outputs a priority-ranked invitation queue for each municipality's health centre coordinator.

Critical principle: The algorithm invites — it never diagnoses. It produces a ranked list of people who should be offered a free lung health check. All clinical decisions are made by humans at the triage visit. The model must remain explainable at every step — no black-box outputs.

STEP 1 — HARD GATES (APPLIED BEFORE ANY SCORING)

These are binary filters. If a patient fails any hard gate, they exit the algorithm entirely — regardless of what other signals exist in their record. No score is calculated. No invitation is generated.

GATE	CONDITION	DATA SOURCE IN KANTA	EXIT REASON
G1 — Age	Patient age < 40 years	Date of birth field (henkilötunnus)	COPD onset rare before 40. Cost-benefit does not support screening younger cohorts.
G2 — Existing diagnosis	ICD-10 code J44.x (COPD) or J45.x (Asthma) present anywhere in record	Diagnosis history, all episodes	Already diagnosed. Route to monitoring programme, not screening invitation.
G3 — Recent screening	Lung health check attended within 24 months	LungFirst procedure code (to be created) or spirometry code within 24 months	Recently screened. Re-evaluate at next biennial cycle.
G4 — Deceased / moved	Death recorded OR municipality of residence outside Pirkanmaa	Population register (Väestörekisteri) link	Not eligible for Pirkanmaa pilot invitation.
G5 — Palliative / terminal	Active palliative care code or terminal diagnosis on record	Care plan codes, oncology records	Screening not clinically appropriate. Do not contact.

STEP 2 — SCORING ENGINE (APPLIED TO ALL WHO PASS GATES)

Five weighted parameters are evaluated. Each parameter queries specific Kanta fields using the keyword library (pages 2–3). Points are additive. Maximum possible score is 7.

PARAMETER	POINTS	PRIMARY KEYWORDS / ICD CODES	LOGIC CONDITION
P1 — Age + Smoking	+2	tupakointi · smoker · ex-smoker · pack years · Nicorette · Champix · varenicline · NRT	Age ≥40 AND (smoking status = current/ex OR NRT/cessation drug prescribed in last 5 years)
P2 — Salbutamol without Dx	+2	Salbutamol · Ventoline · SABA · ATC R03AC02 · no J44 · no J45	≥1 salbutamol prescription dispensed AND no J44/J45 code anywhere in lifetime record
P3 — Antibiotics for LRTI	+1	amoxicillin · doksisykliini · azithromycin · J01 class · J20 · J22 · bronchitis · LRTI	≥2 antibiotic courses in 24 months co-coded with respiratory complaint (J20/J22/R05/R06)
P4 — Vague respiratory visits	+1	yskä · cough · hengenahdistus · dyspnoea · R05 · R06 · J06 · wheeze · sputum	≥2 GP visits in 12 months with respiratory symptom codes and no resolved diagnosis
P5 — No spirometry on record	+1	no spirometry · ei spirometriaa · no FEV1 · no lung function code · spirometry absent	No spirometry procedure code in lifetime record AND patient scores ≥1 on any other parameter

STEP 3 — SCORE BANDING & OUTPUT QUEUE

0–2
No Action

Insufficient signal. Low-risk profile. No invitation generated this cycle.

Re-scored automatically in 6 months. Re-evaluated at biennial cycle.

3
Standard Invite

Moderate signal. Enters Wave 2 invitation queue. Standard appointment slot (20 min).

OmaKanta notification → 3-touch sequence if no response.

4–5
Priority Invite

Strong signal. Enters Wave 1 queue. Priority appointment slot (30 min, sooner date).

OmaKanta + SMS simultaneously. GP receives passive notification.

6–7
Urgent Priority

Maximum signal density. Multiple compounding indicators. Immediate Wave 1 placement.

GP actively notified. Coordinator personally calls if no booking in 7 days.

ICD-10 & ICPC-2 CODES — STRUCTURED FIELD QUERIES

These are queried as exact matches against the diagnosis code fields in Kanta. Split into two categories: **target codes** (presence triggers scoring) and **exclusion codes** (presence stops scoring for that parameter).

J44.0–J44.9
COPD — all subtypes
EXCLUSION — already diagnosed → monitoring track

J45.0–J45.9
Asthma — all subtypes
EXCLUSION from P2 — but keep in algorithm (ACO risk)

J20.0–J20.9
Acute bronchitis
TARGET for P3 — counts as LRTI episode

J22
Unspecified acute LRTI
TARGET for P3 — strongest proxy for undiagnosed COPD exacerbation

J06.9
Acute upper resp. infection unspecified
TARGET for P4 — vague respiratory visit signal

R05
Cough (symptom code)
TARGET for P3 + P4 — unresolved cough is key signal

R06.0–R06.9
Dyspnoea / abnormal breathing
TARGET for P4 — especially R06.0 (dyspnoea) and R06.2 (wheeze)

R09.3
Abnormal sputum
TARGET for P4 — chronic sputum without diagnosis

Z72.0
Tobacco use (current smoker)
TARGET for P1 — primary smoking flag. Also check F17.x

Z87.891
Personal history of tobacco use
TARGET for P1 — ex-smoker flag. Check Kanta social history field

J18.9
Pneumonia unspecified
CONTEXTUAL — single episode: neutral. ≥2 episodes in 3 years: +P3 modifier

J98.0–J98.9
Other respiratory disorders
TARGET for P4 — vague respiratory complaint without resolution

ATC DRUG CODES — PRESCRIPTION DATABASE QUERIES

Bronchodilators — SABA (Parameter P2)
Presence WITHOUT J44/J45 = +2 points

- R03AC02 — Salbutamol
- Ventoline
- Airomir
- Buventol
- R03AC13 — Formoterol (if no Dx)
- R03AC12 — Salmeterol
- terbutaline
- Bricanyl

Smoking Cessation Drugs (Parameter P1)
Confirms smoking history even if not coded

- N07BA01 — Nicotine (NRT)
- Nicorette
- Nicotinell
- N07BA03 — Varenicline
- Champix
- N06AX16 — Bupropion
- Zyban

Antibiotics for Respiratory (Parameter P3)
≥2 courses in 24 months co-coded with J20/J22/R05

- J01CA04 — Amoxicillin
- J01AA02 — Doksisykliini
- J01FA10 — Azithromycin
- J01FA09 — Clarithromycin
- J01MA12 — Levofloxacin
- J01CR02 — Amoxicillin/clavulanate
- J01CE02 — Phenoxymethylpenicillin

Oral Corticosteroids — Short Course (P3 modifier)
Rescue prednisolone = undiagnosed exacerbation proxy

- H02AB07 — Prednisolone
- H02AB06 — Methylprednisolone
- Prednisolon Nycomed
- short course OCS
- oral steroid burst
- kortisoni suun kautta

PROCEDURE CODE ABSENCE CHECK (PARAMETER P5)

The algorithm queries for the **absence** of these codes in the lifetime record. If none are found AND the patient scores ≥1 on any other parameter, P5 adds +1 point.

Spirometry & Lung Function Procedure Codes
Any of these present = P5 does NOT score. None present = +1 if other signals exist.

- TUOHF — Spirometry (Kanta code)
- TUOHG — Spirometry post-bronchodilator
- FEV1 value recorded
- FVC value recorded
- PEF measurement
- peak flow recorded
- microspirometry result
- LungFirst triage code (once issued)
- pulmonology referral letter
- respiratory specialist visit

NLP layer note: The keywords on this page are applied to free-text fields in Kanta — GP notes, referral letters, discharge summaries, nursing records. They do not score independently but act as **confirmatory signals** that raise confidence in structured-field scores, and as **catch signals** for patients whose structured fields are sparse but whose notes reveal clear risk. A natural language processing (NLP) module processes these fields separately and flags them to the structured scoring engine.

FREE TEXT — SYMPTOM VOCABULARY (FINNISH & ENGLISH)

Breathlessness & Airflow

hengenahdistus · dyspnoea · shortness of breath · rasitushengenahdistus · breathless on exertion · hengitys vinkuu · wheeze · wheezing · vinkuva hengitys · hengitysvaikeus · cannot climb stairs · portaat vaikeat · gets breathless walking · kävelyrasitus · pursed lip breathing · tightness in chest · rintakehä ahdas · chest tightness

Cough & Sputum

yskä · krooninen yskä · chronic cough · morning cough · aamuisin yskiminen · productive cough · lima · limaneritys · sputum · phlegm · yskii limaa · mucopurulent · yskiminen rasittaa · coughs every morning

Smoking — Free Text Variants

tupakoitsija · tupakoi · entinen tupakoitsija · lopettanut tupakoinnin · pack years · askivuodet · 20 pack years · heavy smoker · smokes daily · tupakoi päivittäin · pipe smoker · snus · nuuska · ex-smoker 5 years · quit smoking · tupakoinnin lopetus

Functional Decline Phrases

exercise intolerance · rasituksen sietokyvyn lasku · ei jaksa kävellä · fyysinen suorituskyky laskenut · reduced activity · liikkuminen vähentynyt · gets tired easily · väsyy herkästi · stopped exercising · avoids exertion · välttelee rasitusta · hissillä koska hengitys

COMORBIDITY RISK MULTIPLIERS — SCORE MODIFIER (+0.5 ROUNDED UP)

These do not score independently but act as a **0.5-point modifier** — if any of these comorbidities appear alongside a base score of 2.5 or higher, the total rounds up to the next band. This prevents borderline cases from being missed.

Cardiovascular & Metabolic

I50 — heart failure · sydämen vajaatoiminta · I25 — ischaemic heart disease · sepelvaltimotauti · E11 — type 2 diabetes · tyypin 2 diabetes · E66 — obesity BMI >30 · I48 — atrial fibrillation · eteisvärinä

Sleep & Mental Health

G47.3 — sleep apnoea · uniapnea · F32/F33 — depression · masennus · F41 — anxiety · ahdistuneisuus · M81 — osteoporosis · osteoporoosi

OCCUPATIONAL & ENVIRONMENTAL EXPOSURE KEYWORDS

Occupational Exposure — Free Text & Structured Fields

Triggers +0.5 modifier alongside any smoking signal. Queried in occupation field and GP notes.

asbestos · asbestille altistuminen · coal dust · kivihiilipöly · grain dust · viljanpöly · silica · silikoosialtistus · chemical fumes · kemikaalihöyryt · woodworking · puuntyöstö · welding · hitsaus · mining · kaivostyö · construction · rakennustyö · farming · maanviljelys · bakery · leipomo · textile · tekstiiliteollisuus · indoor pollution · biomass smoke · passive smoking · passiivinen tupakointi

⊘ Full Exclusion Keyword List — These Stop Scoring for Specific Parameters

- ⊘ Bronchial asthma confirmed (J45) with active ICS treatment — excludes P2 (salbutamol flag), but patient remains in algorithm for ACO risk monitoring at reduced weight
- ⊘ Active lung cancer (C34.x) or mesothelioma (C45) — full algorithm exit. Route to oncology pathway only.
- ⊘ Pulmonary fibrosis (J84.x) or IPF confirmed — full algorithm exit. Already in specialist care.
- ⊘ Recent spirometry within 24 months with normal result (FEV1/FVC ≥0.75 post-BD) — P5 does not score, but patient stays in algorithm if other parameters still present
- ⊘ Salbutamol prescribed by cardiologist context (cardiac wheeze, heart failure) — requires co-coded I50: downweights P2 by 50% (scores +1 instead of +2)
- ⊘ Anxiety or panic disorder (F41) as sole explanation for dyspnoea — documented in GP notes as "dyspnoea — anxiety-related, not respiratory" — excludes P4 for that visit only

COMPLETE ALGORITHM DECISION TREE — STEP BY STEP

① Ingest patient record from Kanta

Algorithm runs on a **nightly batch cycle** — not real-time. Each night, all updated patient records in the Pirkanmaa Kanta partition are re-evaluated. New prescriptions, new visits, new diagnosis codes trigger re-scoring. Frequency: nightly incremental update, full re-score weekly.

```
# Pseudocode — nightly batch
FOR EACH patient_record IN kanta.pirkanmaa_partition:
    IF record.last_modified > last_run_timestamp:
        queue_for_scoring(patient_record)
RUN full_rescore_weekly() # catches coding updates
```

② Apply hard gates — exit if any gate fails

Evaluated in order. First failure exits immediately — remaining gates not evaluated (efficiency). Gate G1 (age) is evaluated first as it eliminates the largest portion of records fastest.

```
IF age(patient) < 40: EXIT # G1
IF has_code(J44) OR has_code(J45): EXIT → monitoring_track # G2
IF screened_within(24, 'months'): EXIT # G3
IF deceased OR municipality NOT IN pirkanmaa: EXIT # G4
IF has_code(palliative): EXIT # G5
# All gates passed → proceed to scoring
```

③ Calculate composite score — all five parameters evaluated in parallel

Each parameter runs its keyword and code queries simultaneously. Results are summed. Comorbidity modifiers applied last. Score stored with timestamp and parameter breakdown (for explainability).

```
score = 0
# P1 — Age + smoking
IF age >= 40 AND (smoking_coded OR nrt_prescribed OR cessation_drug):
    score += 2
# P2 — Salbutamol without diagnosis
IF salbutamol_prescribed AND NOT (has_code(J44) OR has_code(J45)):
    score += 2
# P3 — Repeated antibiotics for LRTI
IF antibiotic_courses_respiratory(window=24m) >= 2:
    score += 1
# P4 — Vague respiratory visits
IF respiratory_visits_unresolved(window=12m) >= 2:
    score += 1
# P5 — No spirometry on record (only if score > 0)
IF score > 0 AND NOT spirometry_on_record():
    score += 1
# Comorbidity modifier (rounds up if score ends in .5)
IF comorbidity_present() AND score >= 2:
    score += 0.5
score = ceil(score) # round up
```

④ Apply score band → assign to invitation queue

Score is mapped to a band. The output is not just a score — it is a **municipality-specific queue entry** with a calculated invitation week, assigned health centre, and wave designation. Queue is released in batches to prevent capacity overload.

```
IF score <= 2:   → passive_watch(rescore_in='6months')
IF score == 3:   → queue(wave=2, slot='standard_20min')
IF score in [4,5]: → queue(wave=1, slot='priority_30min', notify_gp=True)
IF score >= 6:   → queue(wave=1, slot='urgent_30min', notify_gp=True,
                          coordinator_call_if_no_booking=7)
# Assign to municipality health centre by patient home address
assign_health_centre(patient.municipality, capacity_check=True)
```

5 Invitation sent → response tracked → re-score loop

Once invited, the patient's record is flagged. Non-responders receive the 3-touch sequence (OmaKanta → SMS → letter). Attendance or non-attendance is recorded. **Attended and cleared:** flagged as screened, re-evaluated at biennial cycle. **Attended and positive:** routed to diagnostic pathway, exits algorithm permanently. **Non-attender after 3 touches:** re-invited via biennial campaign. Score continues to update if new signals appear.

BATCHING PROTOCOL — PIRKANMAA CAPACITY MANAGEMENT

Wave 1A **Months 1–2** **Score 7**	~1,500 patients (highest signal density). **Tampere only** — urban pilot. 1 appointment per health centre per day. Nurse trained, microspirometer installed, protocol live. Coordinator reviews daily output.
Wave 1B **Months 3–6** **Score 5–6**	~3,500 patients. All 23 municipalities activated. Rural health centres: 1 appointment every 2 days. Urban centres: 2–3 per day. Pharmacy testing activated in parallel. Mid-pilot review at month 4.
Wave 2 **Months 7–18** **Score 3–4**	~12,000 patients. Steady state: 1–2 appointments per health centre per day. Score 1–2 patients re-scored monthly — any rising to score 3 enter Wave 2 queue immediately. Pharmacy channel continues throughout.
Campaign **Months 19–24** **All scores**	Biennial **Lung Health campaign**. Non-responders from Waves 1–2 re-invited. Walk-in pharmacy testing. Public awareness media. Score 1–2 population offered open invitation. National rollout proposal submitted to THL.

ALGORITHM OUTPUT — WHAT THE HEALTH CENTRE COORDINATOR SEES

Output format: The coordinator's dashboard shows a weekly list for their municipality only. Each entry contains: patient initials + henkilötunnus last 4 digits · score band (not raw score) · which parameters triggered · suggested appointment week · slot type (20 or 30 min) · GP notification status. **Raw scores are never shown to patients** — only the invitation. The full audit trail (which keywords fired, which codes matched) is logged for clinical governance and algorithm validation, accessible to the programme clinical lead only.